drug candidate prioritisation, catalyst design, low-carbon industrial chemistry, advanced battery materials, and biomolecular modelling. As hardware improves and logical qubit counts rise, the field may move from demonstrating isolated wins to delivering repeatable, domain-specific advantages.

Long-term prospects in science and industry

As hardware matures from noisy devices towards fault-tolerant machines with logical qubits and reliable error correction, quantum computers may eventually perform simulations that are beyond the practical reach of even the largest classical supercomputers.

In scientific research, one of the most important long-term outcomes will be more accurate first-principles modelling. Fault-tolerant quantum algorithms such as quantum phase estimation, qubitization, quantum signal processing, and advanced Hamiltonian simulation could enable chemically accurate calculations for larger active spaces than those accessible today. This may transform how researchers study reaction mechanisms, catalytic cycles, photochemical processes, and electron-transfer phenomena. Instead of relying heavily on empirical approximations or small benchmark models, scientists could explore molecular and materials behaviour with higher predictive confidence. This would be particularly valuable in fields such as nitrogen fixation, green ammonia production, carbon dioxide reduction, hydrogen catalysis, and next-generation semiconductor materials.

From an industrial perspective, quantum computing is likely to become part of a broader scientific discovery stack rather than a standalone replacement for classical computing. The future workflow will combine AI for candidate generation, high-performance computing for large-scale screening, quantum processors for high-accuracy treatment of selected quantum subproblems, and automated laboratories for synthesis and testing. In this model, quantum computing contributes most where precision is expensive but scientifically decisive. For example, an AI model may generate thousands of candidate battery electrolytes, classical simulation may filter them, and a quantum algorithm may evaluate the most difficult electronic-structure regions before experimental validation.

The timeline for broad commercial advantage remains uncertain. Near-term devices will continue to be limited by noise, circuit depth, qubit connectivity, and error rates. However, progress in error mitigation, quantum embedding, tensor-network integration, active-space selection, and hybrid quantum-classical algorithms is steadily improving the usefulness of current systems. Over the longer term, advances in logical qubits, scalable control systems, cryogenic engineering, photonic interconnects, and quantum error correction will determine how quickly chemistry-scale quantum advantage becomes practical.

If these technical barriers are overcome, the industrial impact could be considerable. Quantum-enhanced molecular and materials design may reduce development cycles, lower R&D costs, improve catalyst efficiency, accelerate drug discovery, support cleaner energy technologies, and enable materials with properties that are difficult to discover through trial-and-error alone. The most realistic long-term vision is not that quantum computing replaces chemists, materials scientists, or classical simulation, but that it gives them a more powerful instrument for understanding and engineering matter at its most fundamental level.

The most encouraging progress so far has come through hybrid workflows, where quantum systems work alongside classical computing, AI, and laboratory experimentation. Looking ahead, the real promise of quantum computing is not a single dramatic breakthrough, but the gradual expansion of scientific capability—better models, faster discovery cycles, and more informed decisions in areas where precision matters deeply. END

References

- *https://quantumzeitgeist.com/quantum-computing-for-chemistry-and-material-science/*
- *https://www.sciencedirect.com/science/article/pii/S2666386424003837*
- *https://www.mckinsey.com/capabilities/mckinsey-technology/our-insights/solving-chemistrys-toughest-problems-the-quantum-computing-advantage*
- *https://qt.eu/applications/quantum-simulations*
- *https://www.sciencedirect.com/topics/physics-and-astronomy/quantum-simulation*
- *https://www.ddw-online.com/how-quantum-computing-is-revolutionising-drug-development-34423-202504/*
- *https://www.pasqal.com/blog/quantum-computing-and-the-new-era-of-materials-discovery/*
- *https://postquantum.com/quantum-computing/quantum-use-cases-pharma-biotech/*

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a PhD in artificial intelligence and the genetic algorithm. He currently works as a distinguished member of the technical staff (master) and chief architect at Wipro Ltd. This article expresses his views and not of the organisation he works in.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning, and cyber security.